IVF / Fertility Clinic Management System

Doctor

Requirements Specification

Software Engineering – Laboratory Deliverable

Role: Drini Dalipaj / Christian Tasellari(Doctor)

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1. Doctor Role Overview

The Doctor (Drini Dalipaj) is the primary clinical actor in the IVF / Fertility Clinic Management System. The Doctor is the central decision-maker in all treatment episodes, responsible for managing patient care from initial consultation through to IVF cycle completion.

Core Responsibilities:

- Patient Management: View and manage the list of assigned patients, including IVF cycle status and appointment schedules.
- Medical Records: Add and update clinical records including diagnoses, prescriptions, treatment notes, and supporting files.
- Appointment Management: Create, reschedule, and cancel patient appointments with automatic patient notification.
- Laboratory: Request lab tests with clinical instructions, set priority levels, and review results.
- Embryology: Monitor embryo development status, attach clinical notes, and issue transfer/cryopreservation/discard instructions.
- IVF Cycle Management: Review full IVF cycle history and authorize phase transitions.
- Critical Alerts: Receive and acknowledge real-time alerts for critical or abnormal test results.
- Emergency Access: Search and access any patient record in read-only mode during emergencies, subject to mandatory audit logging.

Access Level: The Doctor has read/write access to medical records, lab test orders, embryo clinical notes, IVF cycle phase transitions, and appointments for assigned patients only. The Doctor does not have access to billing, room management, or service catalog configuration (Administrator domain), or to upload laboratory results (Lab Technician domain).

2. Functional Requirements

The following table lists all functional requirements specific to the Doctor role. Each requirement is traceable to one or more user stories.

Req #	Requirement	Priority	Comments / Traceability
FR_D01	The system shall display a list of all patients assigned to the logged-in doctor, showing name, Patient ID, active IVF cycle phase, and next appointment.	1	Supports US_01. List must refresh every 60 seconds.
FR_D02	The system shall allow the doctor to filter the patient list by cycle phase, appointment date, or patient name.	1	Supports US_01.
FR_D03	The system shall allow the doctor to add medical record entries (diagnoses, prescriptions, clinical notes, treatment plan updates) for assigned patients.	1	Supports US_02. Entries are immutable after saving.
FR_D04	The system shall detect and warn the doctor of drug interactions when a prescription entry is submitted.	1	Supports US_02.
FR_D05	The system shall allow the doctor to create, reschedule, and cancel appointments for assigned patients with automatic patient notification.	1	Supports US_03.
FR_D06	The system shall provide a unified calendar view of all the doctor's appointments filterable by date range and patient.	2	Supports US_11.
FR_D07	The system shall allow the doctor to request lab tests from a catalogue, attach clinical instructions, and set priority (Routine / Urgent / Stat).	1	Supports US_04.

FR_D08	The system shall notify the doctor when lab results are available and flag abnormal results as URGENT.	1	Supports US_04, US_07.
FR_D09	The system shall display embryo development data per IVF cycle including cell count, morphology grade, fragmentation, and status.	1	Supports US_05.
FR_D10	The system shall allow the doctor to attach clinical notes to individual embryo records specifying transfer, cryopreservation, or discard instructions.	2	Supports US_09. Discard requires password re-authentication.
FR_D11	The system shall display the full IVF cycle history for each patient including phases, medications, retrieval counts, embryo grades, and outcomes.	1	Supports US_08.
FR_D12	The system shall require doctor authorization before advancing an IVF cycle to the next phase.	1	Supports US_10. Phase skipping is not permitted.
FR_D13	The system shall deliver critical test result alerts to the doctor via in-system notification, email, and SMS within 60 seconds.	1	Supports US_07.
FR_D14	The system shall escalate unacknowledged critical alerts to the on-call supervisor after a configurable window (default 30 minutes).	1	Supports US_07.
FR_D15	The system shall provide a global patient search allowing read-only emergency access to any registered patient with mandatory reason logging.	1	Supports US_12.
FR_D16	The system shall maintain a complete audit trail of all doctor actions including patient access, record entries, and phase authorizations.	1	Required for regulatory compliance.

3. Non-Functional Requirements

Category	Req #	Requirement	Priority
Usability	NFR_D01	The doctor interface shall follow a consistent sidebar navigation with sections: Dashboard, My Patients, Appointments, Lab Tests, Embryology, IVF Cycles, Alerts.	1
Usability	NFR_D02	All doctor forms shall provide inline validation with descriptive error messages before submission.	1
Performance	NFR_D03	The patient list shall load within 2 seconds and refresh every 60 seconds.	1
Performance	NFR_D04	Critical alerts must be delivered within 60 seconds of the triggering event.	1
Security	NFR_D05	Doctor passwords shall be stored using bcrypt hashing.	1
Security	NFR_D06	Authentication shall use JWT tokens with a maximum session duration of 8 hours.	1
Security	NFR_D07	All doctor actions shall be logged in an immutable audit trail with user ID, action, timestamp, and affected record.	1
Security	NFR_D08	Emergency patient access is strictly read-only and generates an automatic audit log and notification to the assigned doctor.	1
Availability	NFR_D09	The system shall be available 99.5% of the time, excluding scheduled maintenance windows.	1
Data Protection	NFR_D10	All patient clinical data shall comply with Albanian Law No. 9887 on Protection of Personal Data and GDPR principles.	1
Data Protection	NFR_D11	Embryo records must be retained for a minimum of 10 years per clinic embryology policy.	1

Data Protection	NFR_D12	Medical record entries are immutable; corrections must be submitted as amendment entries.	1
Portability	NFR_D13	The doctor interface shall be a responsive web application compatible with Chrome, Firefox, Safari, and Edge.	1
Scalability	NFR_D14	The system shall support at least 20 concurrent doctor sessions without performance degradation.	2

4. Doctor User Stories

This section contains 12 user stories for the Doctor role: 6 refined from the original requirements and 6 newly identified during analysis. Each story includes acceptance criteria.

Priorities: High = must-have for MVP; Medium = important for Sprint 3–4.

ID	User Story	Acceptance Criteria	Pri.	Status	Actor
US_01	As a doctor, I want to view the list of patients assigned to me including their IVF cycle status and next appointment so I can plan my daily consultations efficiently.	Patient list shows name, Patient ID, DOB, active IVF cycle phase, and next appointment. Filters work by cycle phase, date, and name. List refreshes every 60 seconds.	High	Refined	Doctor
US_02	As a doctor, I want to add and update medical records for my patients including diagnoses, prescriptions, and treatment notes so the clinical history remains accurate.	Doctor can add Clinical Note, Diagnosis, Prescription, or Treatment Plan Update. Drug interaction warning shown for prescriptions. Entry saved with UTC timestamp and doctor ID.	High	Refined	Doctor
US_03	As a doctor, I want to create, reschedule, and cancel appointments for my patients so visits are coordinated and patients are notified automatically.	Appointments can be created, rescheduled, and cancelled. Conflict detection prevents overbooking. Patients receive automatic notifications for all changes.	High	Refined	Doctor
US_04	As a doctor, I want to request lab tests with clinical instructions and priority level so the lab team processes samples correctly.	Doctor can select tests from catalogue, add instructions, and set Routine/Urgent/Stat priority. Lab Technician is notified. Stat orders trigger SMS within 60 seconds.	High	Refined	Doctor
US_05	As a doctor, I want to view embryo development status including cell count,	Embryo grid shows ID, day, cell count, grade, fragmentation, and status. Day-by-day lab log is	High	Refined	Doctor

	grade, and morphology for IVF patients so I can decide transfer or cryopreservation timing.	accessible. Microscopy images displayed where uploaded.			
US_06	As a doctor, I want to view anonymized donor profile information including compatibility and stock so I can match suitable donors with recipients.	Donor profiles are displayed anonymized per regulations. Compatibility parameters and available stock visible. Doctor cannot see donor identity.	Medium	Refined	Doctor
US_07	As a doctor, I want to receive real-time alerts when a critical test result is uploaded for my patients so I can respond without delay.	Alert delivered via in-system, email, and SMS within 60 seconds. Alert includes patient name, test name, critical value, and direct link. Escalation triggered after 30 minutes if unacknowledged.	High	New	Doctor
US_08	As a doctor, I want to view a patient's complete IVF treatment history across all previous cycles so I can tailor the current cycle protocol.	Full cycle history shows protocols, medications, retrieval counts, embryo grades, and transfer outcomes. Data is read-only and immutable.	High	New	Doctor
US_09	As a doctor, I want to attach structured clinical notes to individual embryo records with transfer, cryopreservation, or discard instructions.	Instruction types: Transfer (Fresh), Transfer (FET), Cryopreserve, Discard, Await. Free-text rationale mandatory. Discard requires password re-authentication. Notes are immutable.	Medium	New	Doctor
US_10	As a doctor, I want to explicitly authorize each IVF cycle phase transition so no phase begins without my sign-off.	Phase transitions require doctor confirmation. Phase skipping not permitted. Prerequisite override requires written justification logged to audit trail.	High	New	Doctor
US_11	As a doctor, I want to view all my appointments in a unified calendar view	Calendar shows upcoming and past appointments. Filterable by date range and patient. No	Medium	New	Doctor

	filterable by date and patient so I can manage my schedule efficiently.	additional navigation required to view full schedule.			
US_12	As a doctor, I want to search for any registered patient by name or ID and access their record in read-only mode in emergencies.	Global search searches all registered patients. Emergency Access modal requires clinical reason (min 10 characters). Access is read-only. Audit log created. Assigned doctor notified.	High	New	Doctor

5. User Scenarios List

The following table summarizes all user scenarios for the Doctor role. Each scenario is expanded in detail in Section 6.

ID	Scenario Name	Description
SC_D01	Doctor logs in	Doctor logs in using credentials and is redirected to the clinical dashboard.
SC_D02	View assigned patient list	Doctor views and filters the list of assigned patients with IVF cycle status.
SC_D03	Open patient profile	Doctor selects a patient and opens their full medical record view.
SC_D04	Add a medical record entry	Doctor adds a diagnosis, prescription, clinical note, or treatment plan update.
SC_D05	Create an appointment	Doctor creates an appointment for an assigned patient.
SC_D06	Reschedule an appointment	Doctor reschedules an existing appointment and patient is notified.
SC_D07	Cancel an appointment	Doctor cancels an appointment with a mandatory reason.
SC_D08	Request a lab test	Doctor requests a lab test with instructions and priority level.
SC_D09	Review lab test results	Doctor reviews uploaded results and marks them as reviewed.
SC_D10	View embryo development status	Doctor views embryo grid and day-by-day development log.
SC_D11	Add clinical note to embryo	Doctor attaches transfer, cryopreservation, or discard instructions to an embryo.

SC_D12	View IVF cycle history	Doctor reviews previous cycle records to inform the current protocol.
SC_D13	Authorize IVF phase transition	Doctor confirms prerequisites and advances the IVF cycle to the next phase.
SC_D14	Receive and acknowledge critical alert	Doctor receives a critical result alert and acknowledges it.
SC_D15	Emergency patient search	Doctor searches for and accesses a non-assigned patient record in an emergency.

6. User Scenarios Extended

Each user scenario is expanded below with step-by-step flows, including both the normal path and exception handling.

SC_D01: Doctor logs in

1. Doctor navigates to the login page.
2. Doctor enters username and password.
3. Doctor clicks 'Login'.
4. System validates credentials.
5. System identifies the user role as Doctor.
6. System redirects to the clinical dashboard.
7. If credentials are invalid: system displays an error and prompts retry.
8. After 5 failed attempts: system locks the account.

SC_D02: View assigned patient list

9. Doctor navigates to 'My Patients' from the dashboard.
10. System retrieves all patients assigned to the logged-in doctor.
11. System displays: full name, Patient ID, DOB, active IVF cycle phase, next appointment.
12. Doctor applies optional filters: by cycle phase, appointment date, or patient name.
13. System refreshes the list to match applied filters.
14. Doctor selects a patient row to open their full profile.
15. System loads the patient's complete medical record view.

SC_D03: Open patient profile

16. Doctor selects a patient from the assigned list.
17. System loads the patient profile: demographics, active IVF cycle, medical record summary.
18. Doctor can navigate to sub-tabs: Medical Records, Appointments, Lab Results, Embryo Status, IVF History.

SC_D04: Add a medical record entry

19. Doctor opens the patient profile.
20. Doctor clicks 'Add Entry' and selects the entry type: Clinical Note, Diagnosis, Prescription, or Treatment Plan Update.
21. System presents the appropriate form fields.

22. Doctor completes all required fields and optionally uploads a supporting file.
23. Doctor submits the entry.
24. System validates required fields and checks for drug interactions if entry type is Prescription.
25. If drug interaction detected: system shows warning modal; doctor must acknowledge before saving.
26. System saves the entry with UTC timestamp and doctor user ID.

SC_D05: Create an appointment

27. Doctor navigates to 'Appointments'.
28. Doctor clicks 'New Appointment'.
29. Doctor selects appointment type (consultation, procedure, follow-up).
30. Doctor chooses date and time slot.
31. System checks for scheduling conflicts.
32. If conflict: system suggests alternative slots; doctor selects one.
33. System creates the appointment and sends notification to the patient.

SC_D06: Reschedule an appointment

34. Doctor selects an existing appointment.
35. Doctor clicks 'Reschedule'.
36. Doctor selects a new date and time slot.
37. System checks for conflicts.
38. System updates the appointment record.
39. System sends automatic reschedule notification to the patient.

SC_D07: Cancel an appointment

40. Doctor selects an appointment.
41. Doctor clicks 'Cancel'.
42. System prompts for a cancellation reason (mandatory).
43. Doctor enters reason and confirms.
44. System cancels the appointment and notifies the patient.

SC_D08: Request a lab test

45. Doctor opens the patient profile and selects 'Request Lab Test'.
46. System displays the test catalogue grouped by category.
47. Doctor selects one or more tests.
48. Doctor enters clinical instructions for the laboratory.

49. Doctor sets the priority level: Routine, Urgent, or Stat.
50. Doctor submits the order.
51. System creates a lab test order with status 'Pending'.
52. System notifies the designated Lab Technician.

SC_D09: Review lab test results

53. Doctor receives notification that results are available.
54. Doctor navigates to the patient's lab results tab.
55. System displays results with reference ranges; out-of-range values are flagged.
56. Doctor reviews the result report.
57. Doctor clicks 'Mark as Reviewed'; system unlocks patient portal view of the result.

SC_D10: View embryo development status

58. Doctor opens the patient's IVF profile and navigates to 'Embryo Status'.
59. System displays all embryos: ID, day, cell count, morphology grade, fragmentation, status.
60. Doctor selects an individual embryo to expand its full development timeline.
61. System shows the day-by-day log of Lab Technician observations and any microscopy images.
62. Doctor assesses embryo quality based on the development log.

SC_D11: Add clinical note to embryo

63. Doctor clicks 'Add Clinical Note' for a selected embryo.
64. Doctor selects instruction type: Transfer (Fresh), Transfer (FET), Cryopreserve, Discard, or Await Further Observation.
65. Doctor enters a free-text clinical rationale (mandatory).
66. If 'Discard': system displays a secondary confirmation modal requiring password re-entry.
67. Doctor confirms and submits.
68. System saves the note and highlights the instruction on the Lab Technician's embryology dashboard.

SC_D12: View IVF cycle history

69. Doctor opens the patient's IVF profile.
70. System displays the active cycle dashboard and a collapsible list of all previous cycles.
71. Doctor selects a past cycle; system expands: phases, dates, hormone trends, retrieval counts, embryo grades, transfer details, and final outcome.

72. Doctor uses historical data to inform adjustments to the current cycle protocol.

SC_D13: Authorize IVF phase transition

73. Doctor confirms all prerequisites for the next phase are complete.
74. Doctor clicks 'Advance to Next Phase'.
75. System presents a phase-transition confirmation modal: current phase, next phase, any outstanding prerequisite warnings.
76. Doctor reviews the modal, adds an optional clinical note, and confirms.
77. System updates the cycle phase with UTC timestamp and doctor user ID.
78. System sends a phase-change notification to the patient.
79. System generates task notifications for the Lab Technician for the new phase.

SC_D14: Receive and acknowledge critical alert

80. System detects a critical result (auto-evaluation or Lab Technician flag).
81. System delivers alert via in-system notification, email, and SMS within 60 seconds.
82. Alert includes: patient name, Patient ID, test name, critical value, reference range, and direct link.
83. Doctor opens the patient record and reviews the full result.
84. Doctor clicks 'Acknowledge & Review'.
85. System timestamps the acknowledgement. If not acknowledged within 30 minutes: system escalates to on-call supervisor.

SC_D15: Emergency patient search

86. Doctor types a patient name or ID into the global search bar.
87. System searches all registered patient records.
88. System returns results showing: patient name, Patient ID, DOB (partially masked), assigned doctor.
89. Doctor selects the target patient.
90. System presents the Emergency Access modal requiring a clinical reason (minimum 10 characters).
91. Doctor enters reason and confirms.
92. System grants read-only access, creates an audit log entry, and notifies the patient's assigned doctor.

7. Use Cases – Doctor User Flows

Each use case describes the full user flow for the Doctor, following the structured format from the course case study.

7.1 UC_D01: View Assigned Patient List

UC Name	View Assigned Patient List
UC ID	UC_D01
Actors	Doctor (primary)
Description	The doctor views and filters the list of patients currently assigned to them, including each patient's IVF cycle status and next appointment date.
Preconditions	Doctor is authenticated. At least one patient has been assigned by an Administrator. Patient records contain basic demographic and cycle information.
Main Flow	1. Doctor navigates to 'My Patients'. 2. System retrieves all assigned patients. 3. System displays: full name, Patient ID, DOB, IVF cycle phase, next appointment. 4. Doctor applies optional filters. 5. System refreshes list. 6. Doctor selects a patient to open their profile. 7. System loads the full patient medical record view.
Alternative / Exception Flow	2a. No patients assigned: informational banner shown. 4a. Filter returns no results: reset button offered. Emergency: doctor uses global search (UC_D07) for non-assigned patients.
What Can Go Wrong	Stale data if list is not refreshed. Mitigation: 60-second auto-refresh. Doctor may filter too narrowly. Mitigation: reset filter button always visible.
Postconditions	Doctor has a current filtered view. Selected profile is fully loaded. No data modified.
Restrictions & Constraints	Doctor may only view patients assigned by Administrator. List must not cache data for more than 60 seconds. Out-of-assignment access is read-only and audit-logged.
Related User Stories	US_01, US_12
Dependencies	None — patient assignment is performed by the Administrator.

7.2 UC_D02: Add / Update Medical Record

UC Name	Add / Update Medical Record
UC ID	UC_D02
Actors	Doctor (primary)

Description	The doctor adds new entries to a patient's medical record, including diagnoses, prescriptions, clinical notes, and treatment plan updates.
Preconditions	Doctor is authenticated. Patient is assigned to this doctor. A patient record exists in the system.
Main Flow	1. Doctor opens patient profile. 2. System displays full medical record chronologically. 3. Doctor clicks 'Add Entry' and selects type. 4. System presents form fields. 5. Doctor completes fields and optionally uploads files. 6. Doctor submits. 7. System validates and checks drug interactions if Prescription. 8. System saves with UTC timestamp and doctor ID. 9. System displays updated record with new entry highlighted.
Alternative / Exception Flow	3a. No record exists: system offers to create initial record shell. 7a. Validation error: fields highlighted; doctor corrects. 7b. Drug interaction: warning modal shown; doctor must acknowledge before saving.
What Can Go Wrong	Incorrect data entry. Mitigation: mandatory field validation. File too large. Mitigation: 20 MB limit enforced. Drug interaction missed. Mitigation: automatic interaction check on every prescription.
Postconditions	Record updated with new timestamped entry. All prior entries intact. Patient portal reflects updated summary.
Restrictions & Constraints	Only the assigned doctor may add entries. Entries cannot be deleted; corrections via separate Amendment entries. File uploads max 20 MB. Must comply with GDPR and HIPAA.
Related User Stories	US_02, US_08
Dependencies	UC_D01 — patient must be in the doctor's assigned list.

7.3 UC_D03: Create and Manage Appointments

UC Name	Create and Manage Appointments
UC ID	UC_D03
Actors	Doctor (primary), Patient (notified)
Description	The doctor creates, reschedules, or cancels appointments for assigned patients, with automatic patient notification for all changes.
Preconditions	Doctor is authenticated. Patient is assigned to this doctor.
Main Flow	Creating: 1. Doctor navigates to Appointments. 2. Clicks 'New Appointment'. 3. Selects type, date, and time slot. 4. System checks for conflicts. 5. Creates appointment; notifies patient. Rescheduling: 1. Doctor selects appointment. 2. Chooses new slot. 3. System checks conflicts; updates and notifies. Cancelling: 1. Doctor selects appointment. 2. Enters mandatory reason. 3. System cancels and notifies patient.
Alternative / Exception Flow	Conflict detected: system suggests alternative slots. Patient has no upcoming appointments: empty state shown with 'New Appointment' prompt.

What Can Go Wrong	Notification fails. Mitigation: delivery logged; administrator can resend. Conflict not caught. Mitigation: real-time conflict detection at submission.
Postconditions	Appointment created, rescheduled, or cancelled. Patient notified. Event logged in audit trail.
Restrictions & Constraints	Appointments only within doctor's configured availability. Cancellation requires a mandatory reason. Notifications must be sent within 1 minute.
Related User Stories	US_03, US_11
Dependencies	UC_D01 — patient must be assigned to the doctor.

7.4 UC_D04: Request Lab Test and Review Results

UC Name	Request Lab Test and Review Results
UC ID	UC_D04
Actors	Doctor (initiator), Lab Technician (fulfiller), Patient (recipient)
Description	The doctor requests lab tests for a patient from a configured catalogue, attaches clinical instructions and priority, and reviews results once uploaded.
Preconditions	Doctor is authenticated. Patient is assigned to this doctor. At least one lab test type exists in the service catalogue.
Main Flow	1. Doctor opens patient profile > 'Request Lab Test'. 2. System shows test catalogue by category. 3. Doctor selects tests. 4. Doctor enters clinical instructions. 5. Doctor sets priority (Routine / Urgent / Stat). 6. Doctor submits order. 7. System creates order with status 'Pending'. 8. System notifies Lab Technician. 9. Lab Technician uploads results. 10. System auto-evaluates against reference ranges. 11. System notifies doctor; flags abnormal as URGENT. 12. Doctor reviews and marks as 'Reviewed'. 13. System unlocks patient portal view of result.
Alternative / Exception Flow	Test not in catalogue: doctor contacts Administrator. Stat priority: immediate SMS to on-call Lab Technician. Abnormal result: escalated notification before standard. Cancel: doctor can cancel Pending order before Lab Technician begins.
What Can Go Wrong	Lab Technician misses Stat order. Mitigation: SMS alert within 60 seconds. Doctor misses URGENT result. Mitigation: escalation if not acknowledged within 30 minutes.
Postconditions	Lab order created and linked. Results permanently attached to record. Doctor has reviewed and acknowledged. Patient can view result (subject to review gate).
Restrictions & Constraints	Only assigned doctors may create lab orders. Cancelled orders retained in audit log. Stat orders must trigger within 60 seconds.
Related User Stories	US_04, US_07

Dependencies	UC_D01 — patient must be assigned. Lab Technician role must upload results (UC_LT01).

7.5 UC_D05: View Embryo Status and Add Clinical Notes

UC Name	View Embryo Status and Add Clinical Notes
UC ID	UC_D05
Actors	Doctor (primary), Lab Technician (data provider)
Description	The doctor reviews embryo development data and attaches structured clinical notes specifying transfer, cryopreservation, or discard instructions.
Preconditions	Doctor is authenticated. Patient has an active IVF cycle with at least one fertilized embryo. Lab Technician has entered at least one development observation.
Main Flow	1. Doctor opens patient IVF profile > 'Embryo Status'. 2. System shows embryo grid: ID, day, cell count, grade, fragmentation, status. 3. Doctor selects embryo to expand timeline. 4. System shows day-by-day log and microscopy images. 5. Doctor clicks 'Add Clinical Note'. 6. Doctor selects instruction type. 7. Doctor enters mandatory rationale. 8. For Discard: password re-authentication required. 9. Doctor confirms. 10. System saves note and updates Lab Technician's dashboard.
Alternative / Exception Flow	No embryo data: 'Awaiting first lab entry' shown. Arrested development: red warning indicator; doctor prompted to add note. Doctor cancels Discard: no note saved.
What Can Go Wrong	Doctor issues incorrect instruction. Mitigation: notes are immutable; addendum entries required. Discard without rationale. Mitigation: rationale is mandatory.
Postconditions	Clinical note permanently attached to embryo record. Lab Technician dashboard updated. Discard confirmations double-logged.
Restrictions & Constraints	Only assigned doctor may add notes. Notes cannot be deleted; addendum entries only. Discard requires written rationale and password re-authentication. Embryo records retained minimum 10 years.
Related User Stories	US_05, US_09
Dependencies	Lab Technician must have entered embryo development observations.

7.6 UC_D06: View IVF Cycle History and Authorize Phase Transitions

UC Name	View IVF Cycle History and Authorize Phase Transitions
UC ID	UC_D06

Actors	Doctor (primary), Patient (notified), Lab Technician (notified)
Description	The doctor reviews the full IVF treatment history across all cycles and authorizes transitions between IVF cycle phases.
Preconditions	Doctor is authenticated and assigned to the patient. Patient has at least one IVF cycle recorded.
Main Flow	1. Doctor opens patient IVF profile. 2. System shows active cycle dashboard and collapsible previous cycle list. 3. Doctor reviews past cycles: phases, dates, hormone trends, retrieval counts, embryo grades, transfer outcomes. 4. For active cycle: doctor confirms prerequisites are complete. 5. Doctor clicks 'Advance to Next Phase'. 6. System shows transition modal: current phase, next phase, outstanding warnings. 7. Doctor reviews, adds optional note, and confirms. 8. System updates phase with UTC timestamp and doctor ID. 9. System notifies patient and Lab Technician.
Alternative / Exception Flow	No past cycles: 'First cycle' messaging shown. Prerequisite not met: system highlights item; doctor must provide written justification to override (audit-logged). Cycle cancellation: doctor selects reason code and free-text note; patient notified immediately.
What Can Go Wrong	Phase advanced prematurely. Mitigation: prerequisite checklist enforced. Patient not notified. Mitigation: automatic notification within 1 minute.
Postconditions	Phase updated and timestamped. Patient and Lab Technician notified. All history remains immutable.
Restrictions & Constraints	Only assigned doctor may authorize transitions. Phases must advance in defined clinical sequence; skipping not permitted. Cancellation requires reason code and minimum 20-character note.
Related User Stories	US_08, US_10
Dependencies	UC_D04 — required test results must be uploaded before phase transition prerequisites can be met.

7.7 UC_D07: Emergency Patient Search

UC Name	Emergency Patient Search
UC ID	UC_D07
Actors	Doctor (requesting), System (access control & audit)
Description	The doctor accesses the record of a patient not on their assigned list in an emergency, subject to mandatory reason-logging and read-only access.
Preconditions	Doctor is authenticated. Target patient is registered. Patient is not on the doctor's assigned list.
Main Flow	1. Doctor types patient name or ID into the global search bar. 2. System searches all registered patients. 3. System returns: name, Patient ID, DOB (partially masked), assigned doctor. 4. Doctor selects patient. 5. System

	presents Emergency Access modal: reason required (min 10 characters). 6. Doctor enters reason and confirms. 7. System grants read-only access for the current session. 8. System creates audit log entry: requesting doctor ID, patient ID, timestamp, reason. 9. System notifies the patient's assigned doctor.
Alternative / Exception Flow	No patient found: 'No matching record found' shown. Doctor cancels reason modal: access blocked; attempt logged as 'Declined by User'. Fewer than 10 characters: submission rejected.
What Can Go Wrong	Access misused. Mitigation: full audit trail included in weekly governance report. Doctor forgets to note reason. Mitigation: minimum 10-character requirement enforced.
Postconditions	Doctor has read-only access for the current session. Audit log entry created. Assigned doctor notified.
Restrictions & Constraints	Emergency access is strictly read-only. Reason field mandatory, minimum 10 characters. Feature cannot be disabled by any individual user. Access revoked at session end.
Related User Stories	US_12
Dependencies	None — only requires that the patient is registered in the system.

8. Use Case Diagram Description

The Use Case Diagram describes the Doctor actor and all use cases they interact with, following UML notation with «include» and «extend» relationships.

Actor:

- Doctor (Drini Dalipaj)

Primary Use Cases:

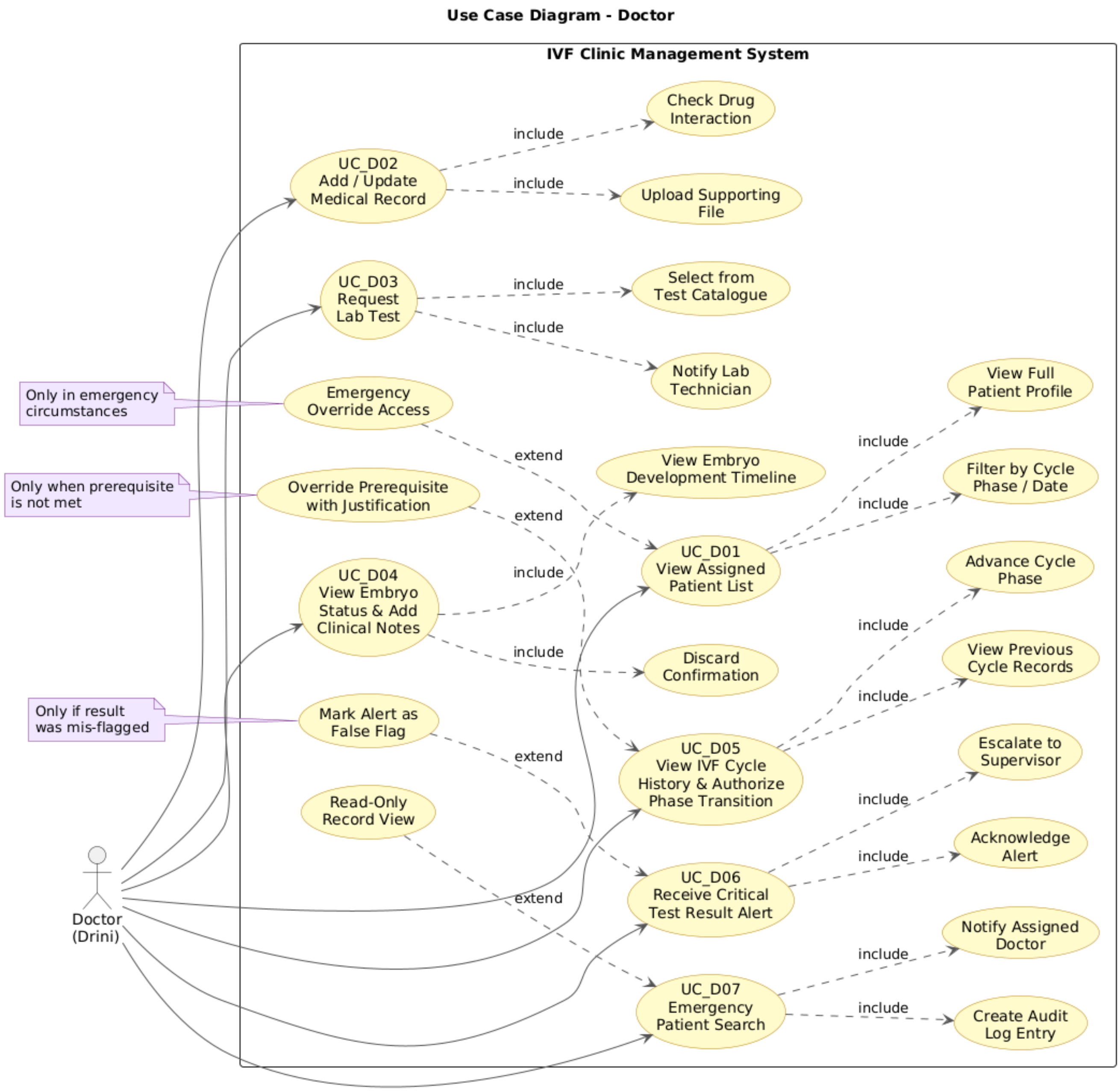
- View Assigned Patient List
- Add / Update Medical Record
- Create and Manage Appointments
- Request Lab Test & Review Results
- View Embryo Status & Add Clinical Notes
- View IVF Cycle History & Authorize Phase Transition
- Receive Critical Test Result Alert
- Emergency Patient Search

«include» Relationships:

- View Assigned Patient List «include» Filter by Cycle Phase
- View Assigned Patient List «include» View Full Patient Profile
- Add / Update Medical Record «include» Check Drug Interaction
- Add / Update Medical Record «include» Upload Supporting File
- Request Lab Test «include» Select from Test Catalogue
- Request Lab Test «include» Notify Lab Technician
- View Embryo Status «include» View Embryo Development Timeline
- View Embryo Status «include» Discard Confirmation
- View IVF Cycle History «include» View Previous Cycle Records
- View IVF Cycle History «include» Advance Cycle Phase
- Receive Critical Alert «include» Acknowledge Alert
- Receive Critical Alert «include» Escalate to Supervisor
- Emergency Patient Search «include» Create Audit Log Entry
- Emergency Patient Search «include» Notify Assigned Doctor

«extend» Relationships:

- View Assigned Patient List «extend» Emergency Override Access — only in emergency circumstances
- View IVF Cycle History «extend» Override Prerequisite with Justification — only when prerequisite not met
- Receive Critical Alert «extend» Mark Alert as False Flag — only if result was mis-flagged

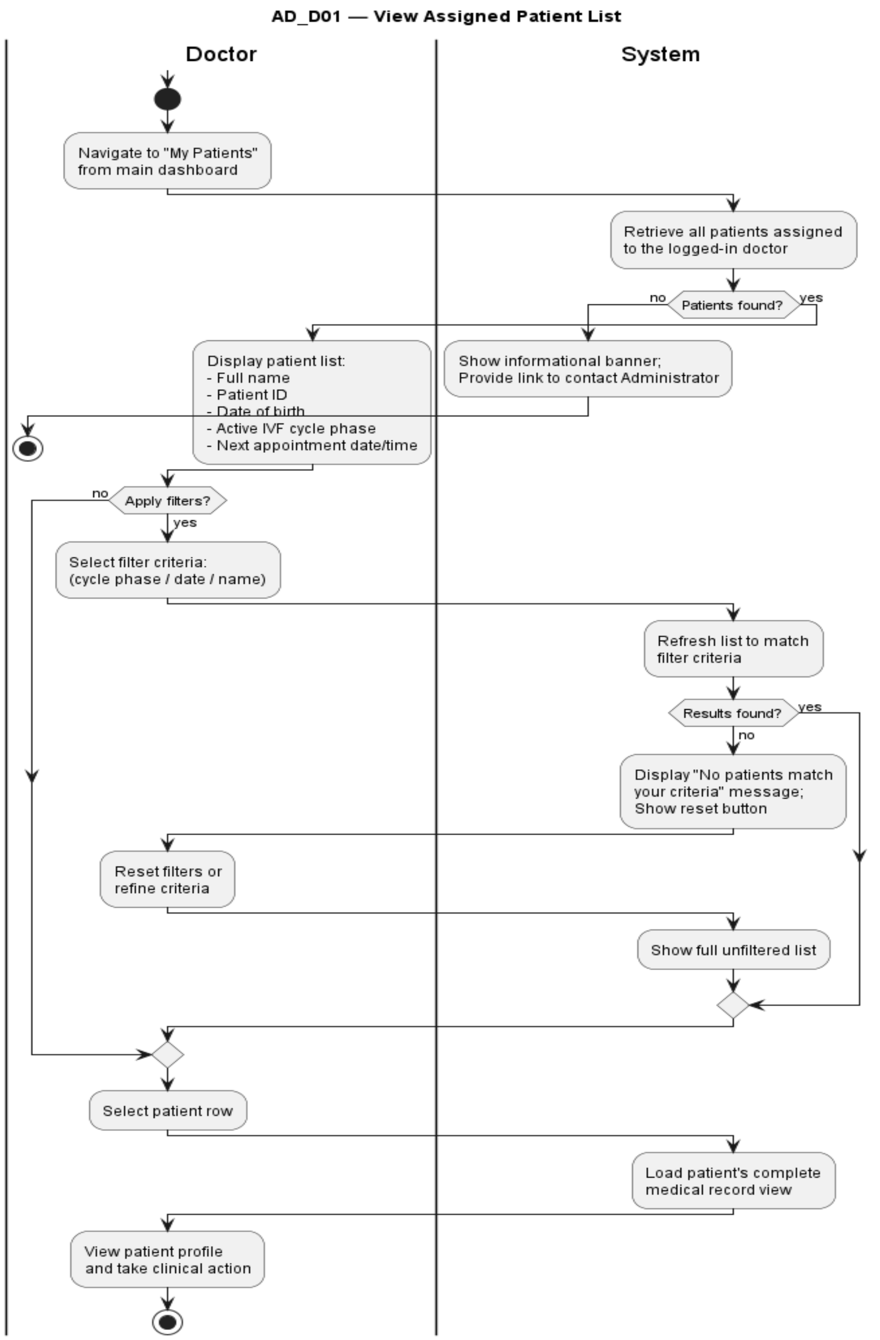


Use Case Diagram - Doctor
IVF Clinic Management System
Doctor (Drini)
UC_D02 Add / Update Medical Record
Check Drug Interaction
Upload Supporting File
UC_D03 Request Lab Test
Select from Test Catalogue
Notify Lab Technician
Only in emergency circumstances
Emergency Override Access
Only when prerequisite is not met
Override Prerequisite with Justification
View Embryo Development Timeline
UC_D01 View Assigned Patient List
View Full Patient Profile
Filter by Cycle Phase / Date
UC_D04 View Embryo Status & Add Clinical Notes
Discard Confirmation
Advance Cycle Phase
View Previous Cycle Records
Only if result was mis-flagged
Mark Alert as False Flag
UC_D05 View IVF Cycle History & Authorize Phase Transition
Read-Only Record View
Escalate to Supervisor
UC_D06 Receive Critical Test Result Alert
Acknowledge Alert
UC_D07 Emergency Patient Search
Notify Assigned Doctor
Create Audit Log Entry
include
extend

9. Activity Diagrams

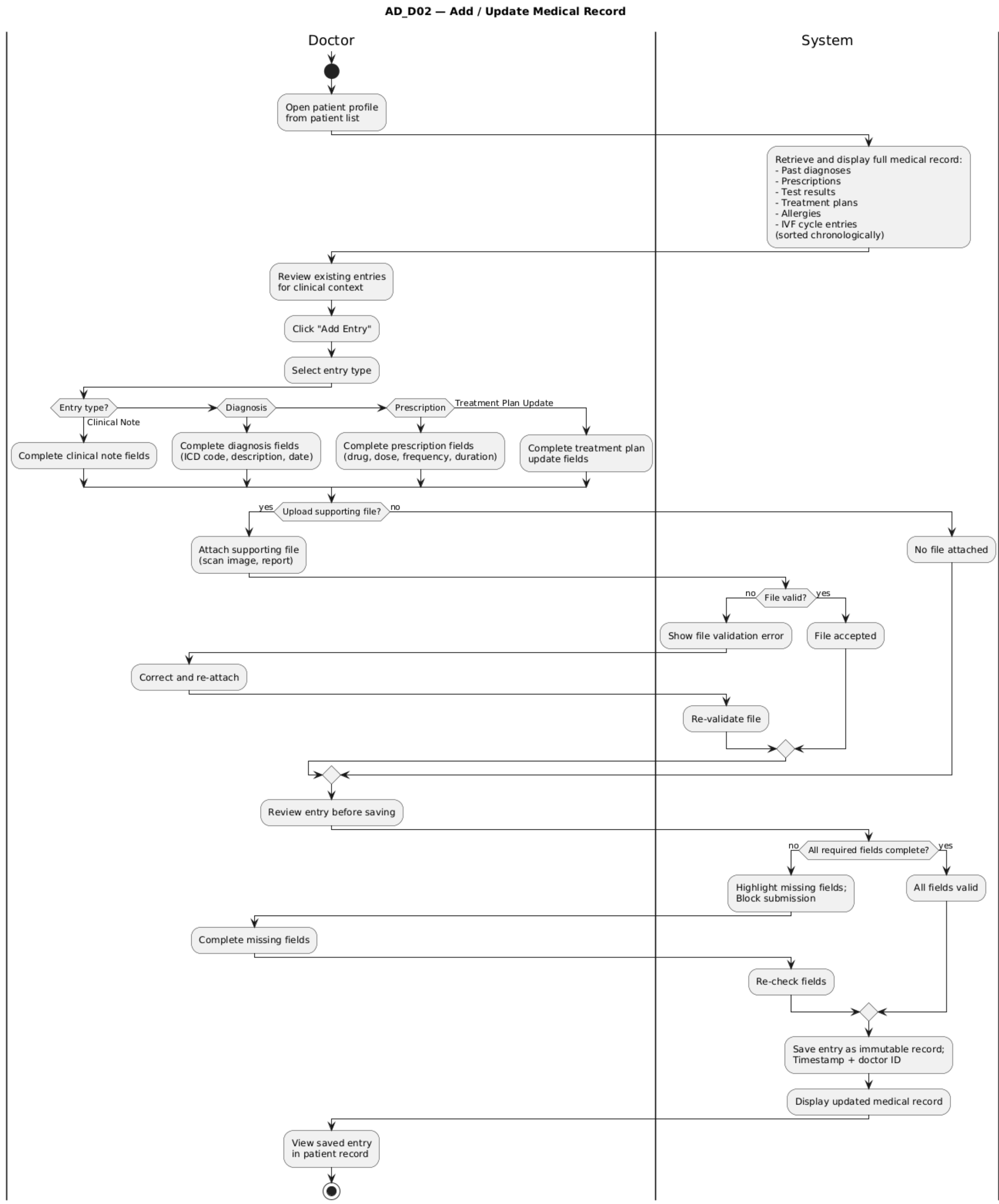
This section contains the activity diagrams for each Doctor use case. Insert the corresponding PNG image in the placeholder below each title.

9.1 UC_D01: View Assigned Patient List



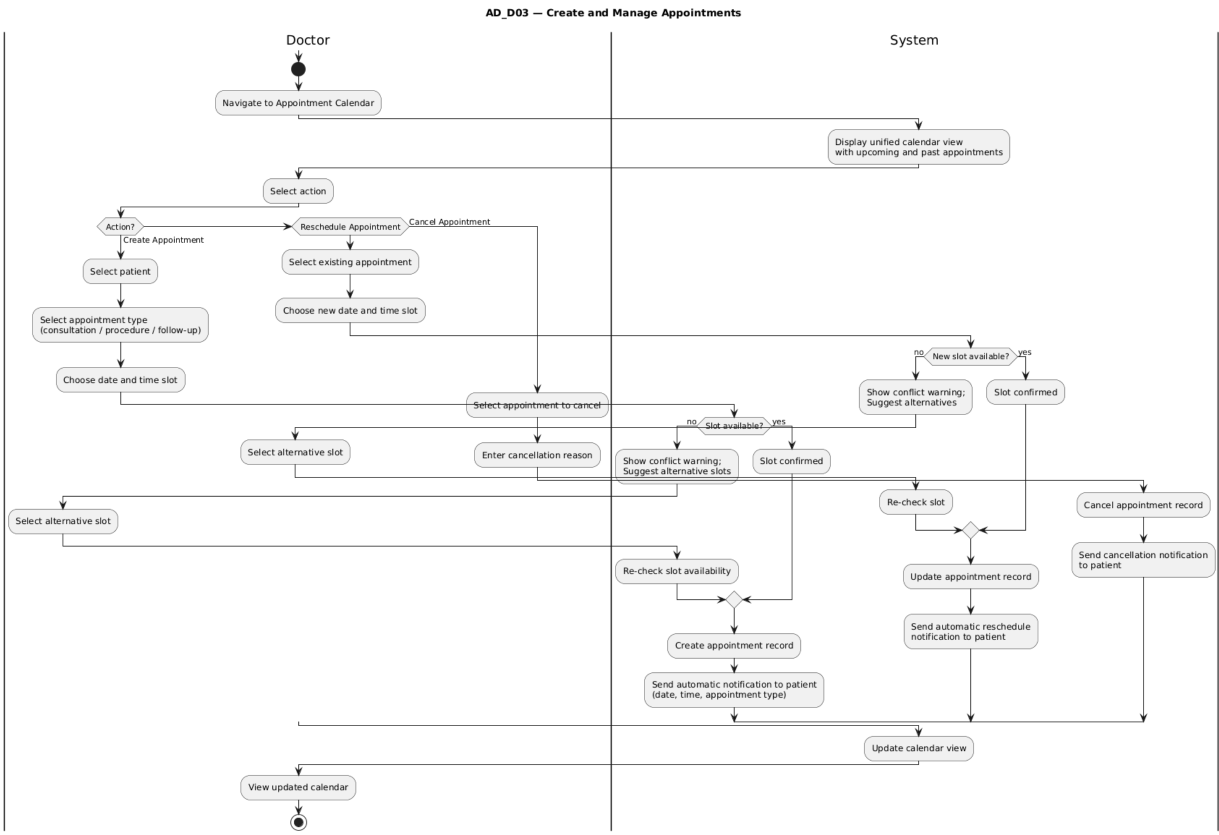
Activity diagram for UC_D01: View Assigned Patient List

9.2 UC_D02: Add / Update Medical Record



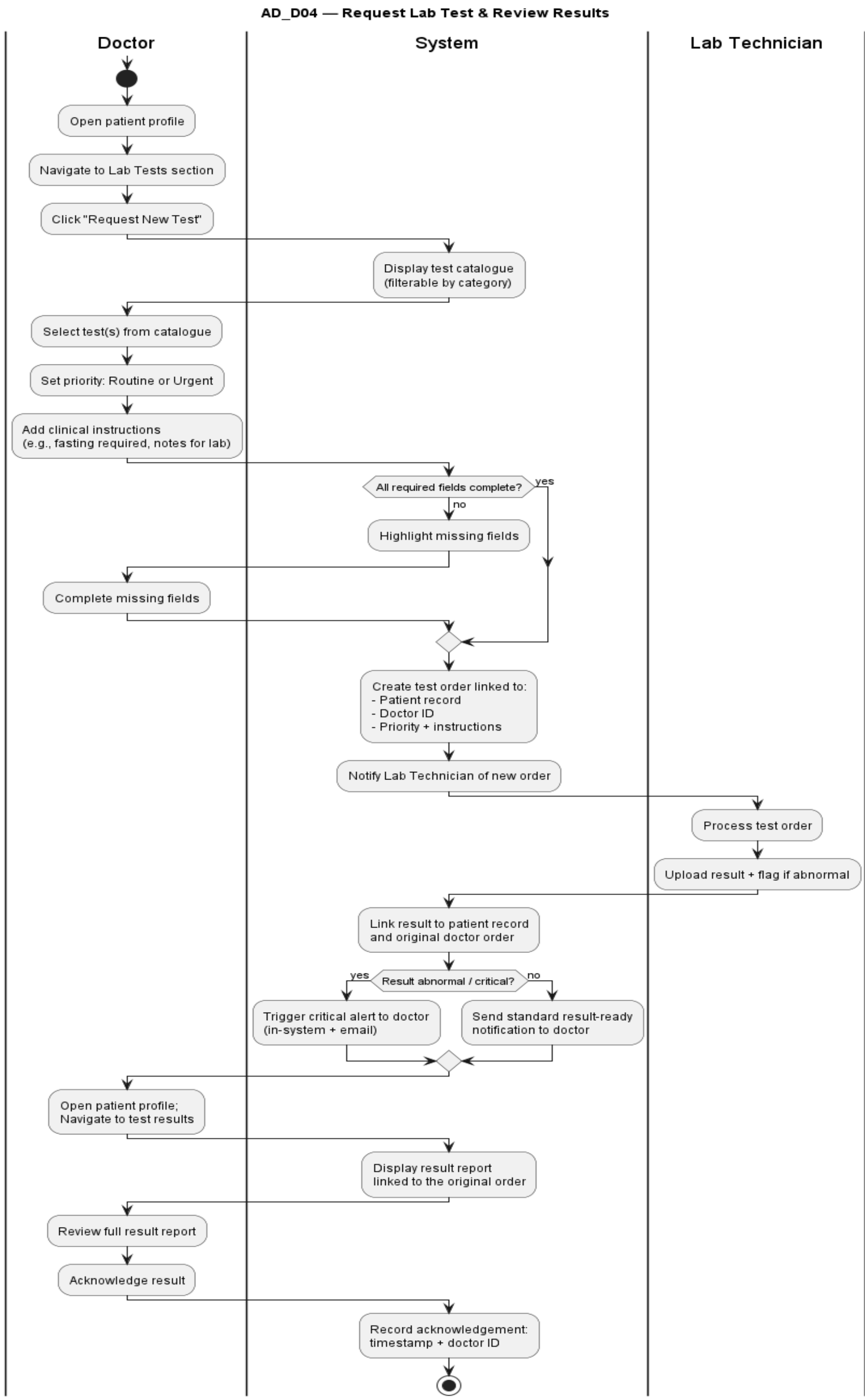
Activity diagram for UC_D02: Add / Update Medical Record

9.3 UC_D03: Create and Manage Appointments



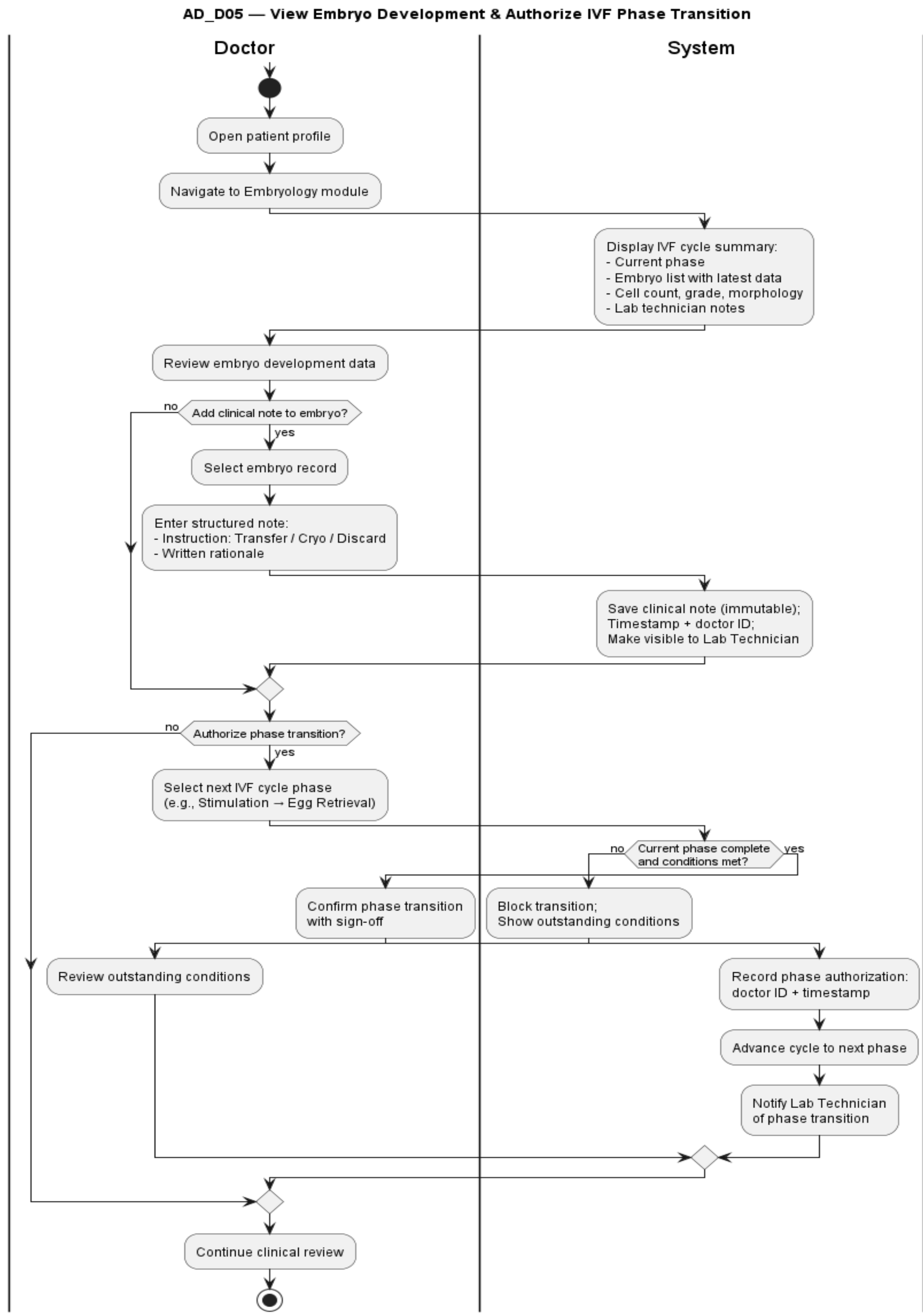
Activity diagram for UC_D03: Create and Manage Appointments

9.4 UC_D04: Request Lab Test and Review Results



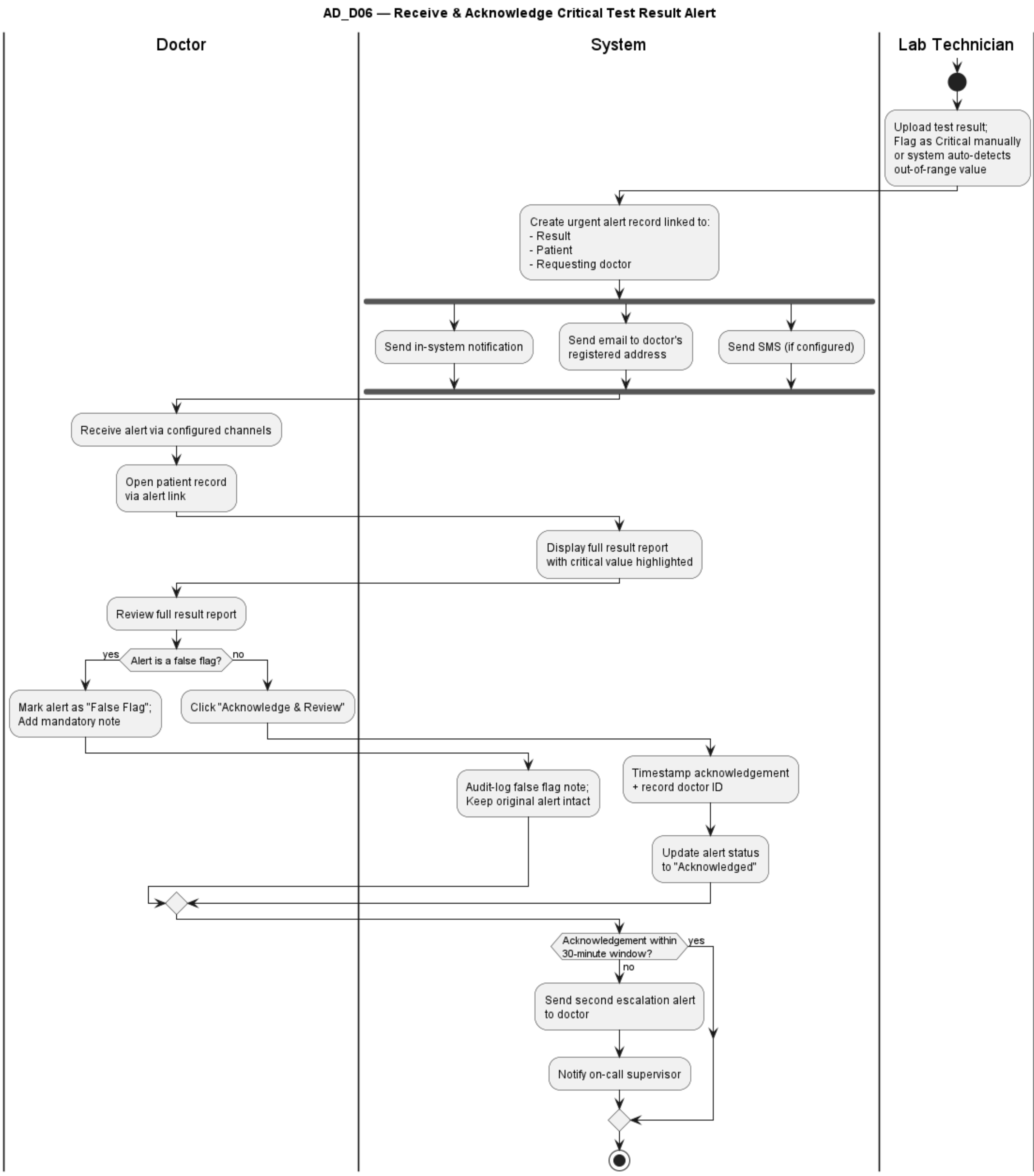
Activity diagram for UC_D04: Request Lab Test and Review Results

9.5 UC_D05: View Embryo Status and Add Clinical Notes



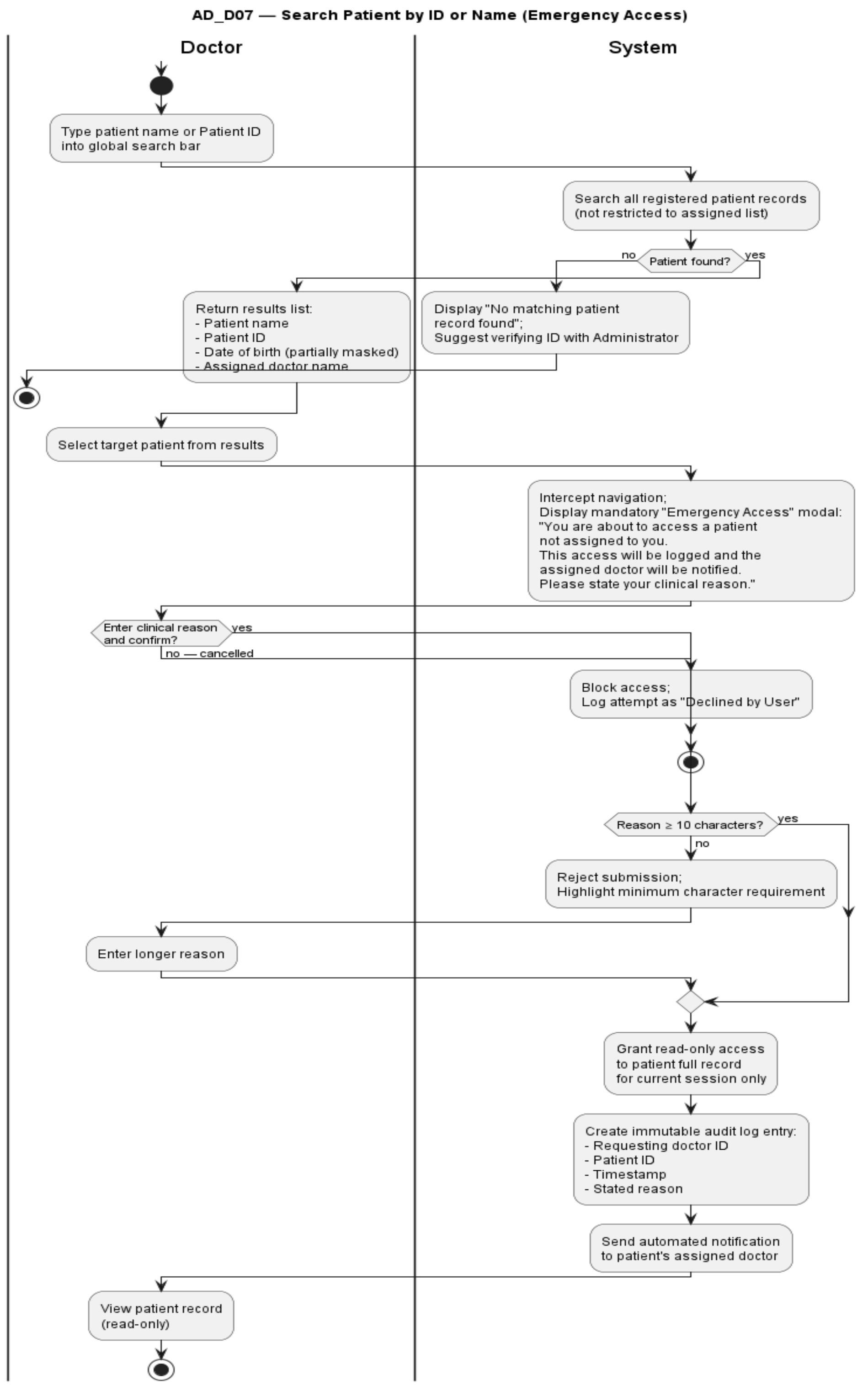
Activity diagram for UC_D05: View Embryo Status and Add Clinical Notes

9.6 UC_D06: View IVF Cycle History and Authorize Phase Transition



Activity diagram for UC_D06: View IVF Cycle History and Authorize Phase Transition

9.7 UC_D07: Emergency Patient Search

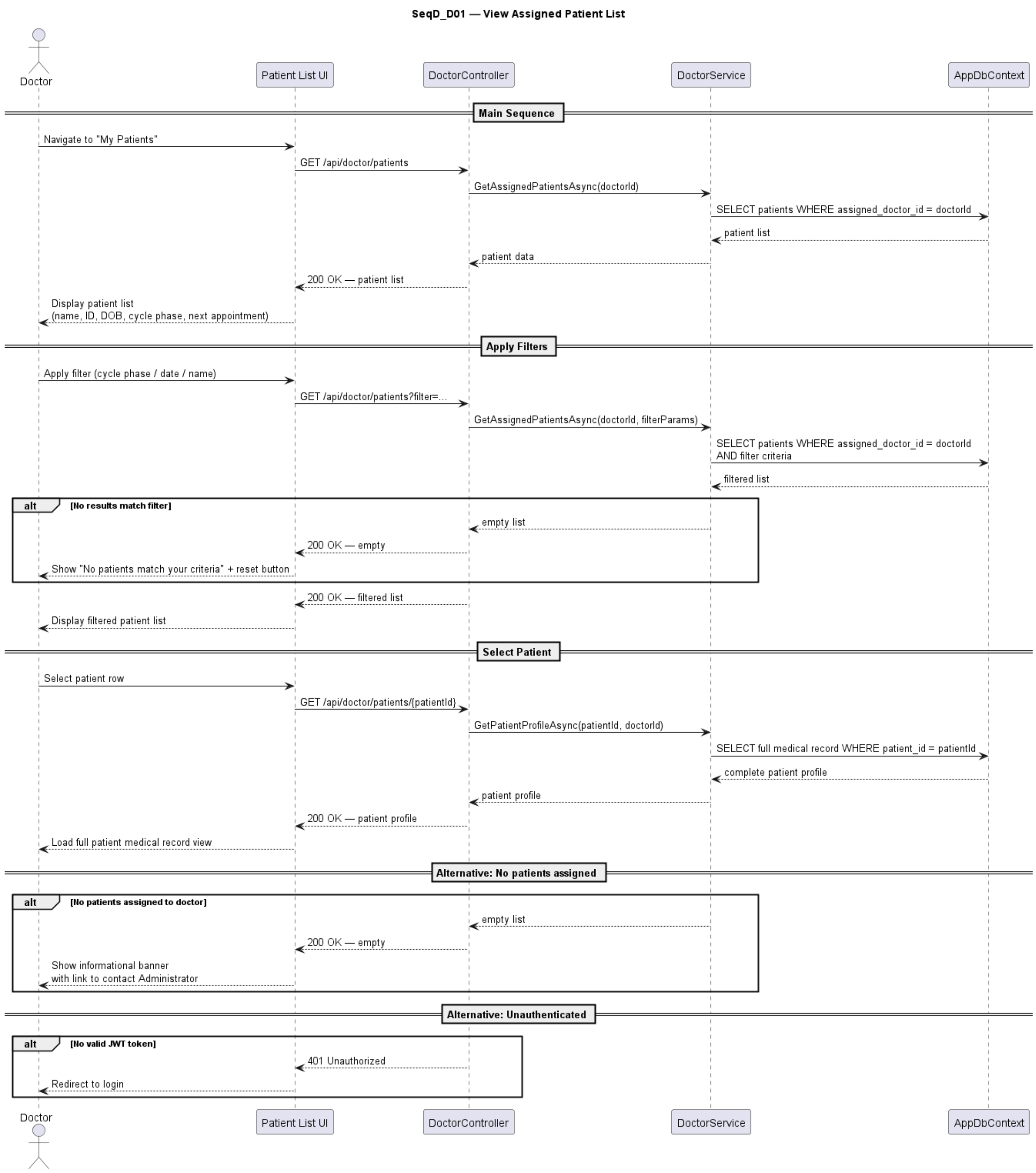


Activity diagram for UC_D07: Emergency Patient Search

10. Sequence Diagrams

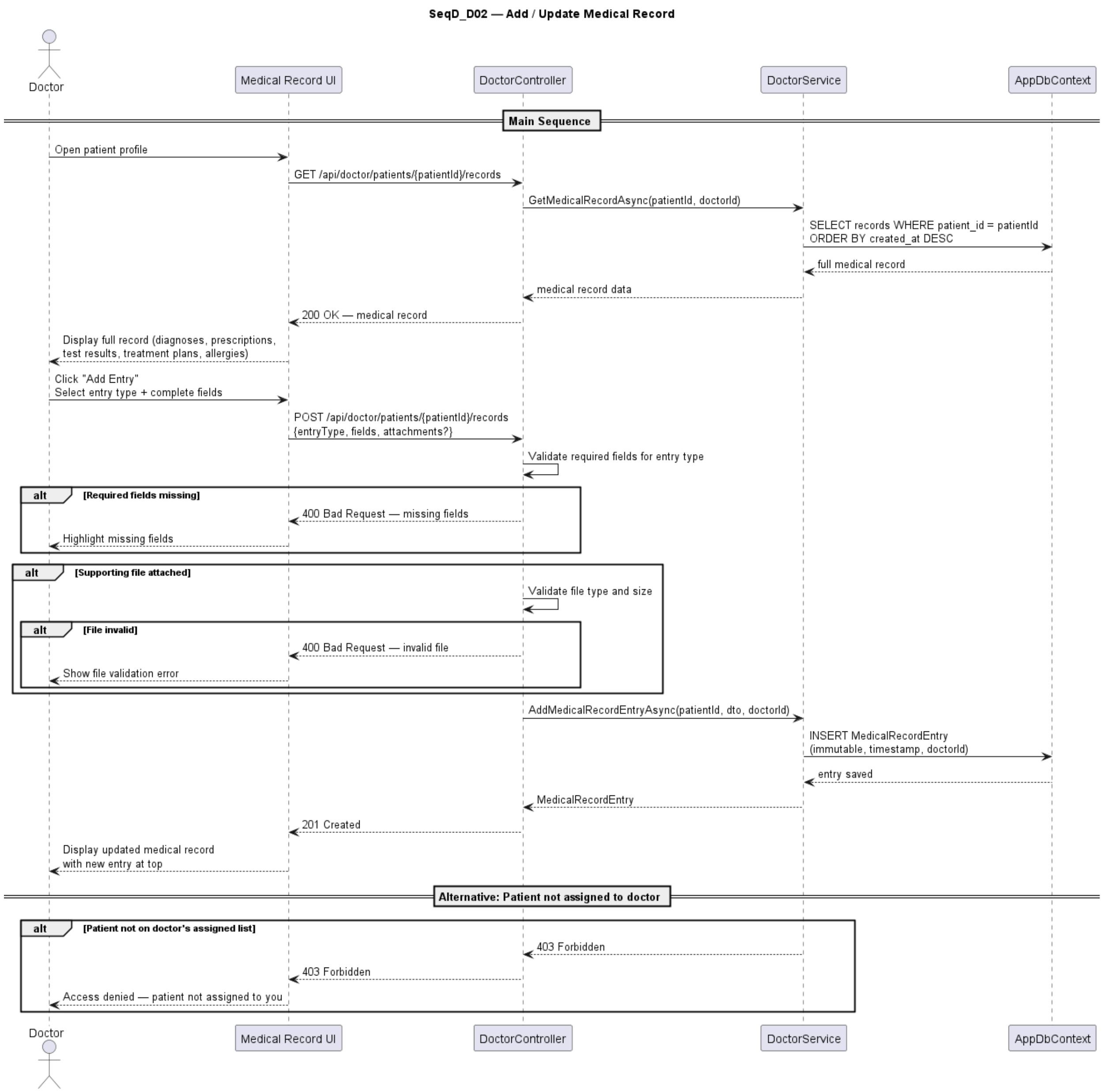
This section contains the sequence diagrams for each Doctor use case. Insert the corresponding PNG image in the placeholder below each title.

10.1 UC_D01: View Assigned Patient List



Sequence diagram for UC_D01: View Assigned Patient List

10.2 UC_D02: Add / Update Medical Record

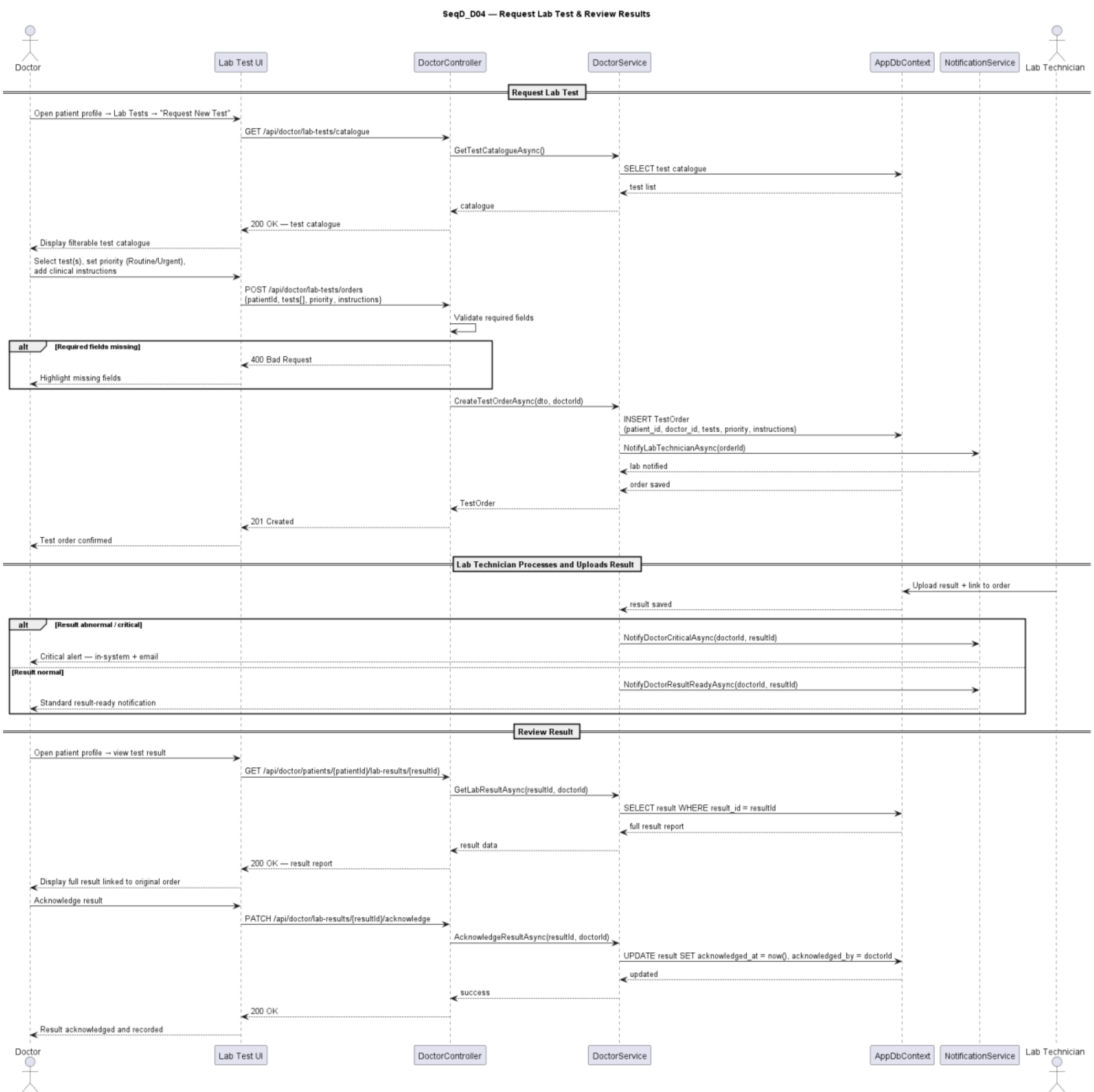


Sequence diagram for UC_D02: Add / Update Medical Record

10.3 UC_D03: Create and Manage Appointments

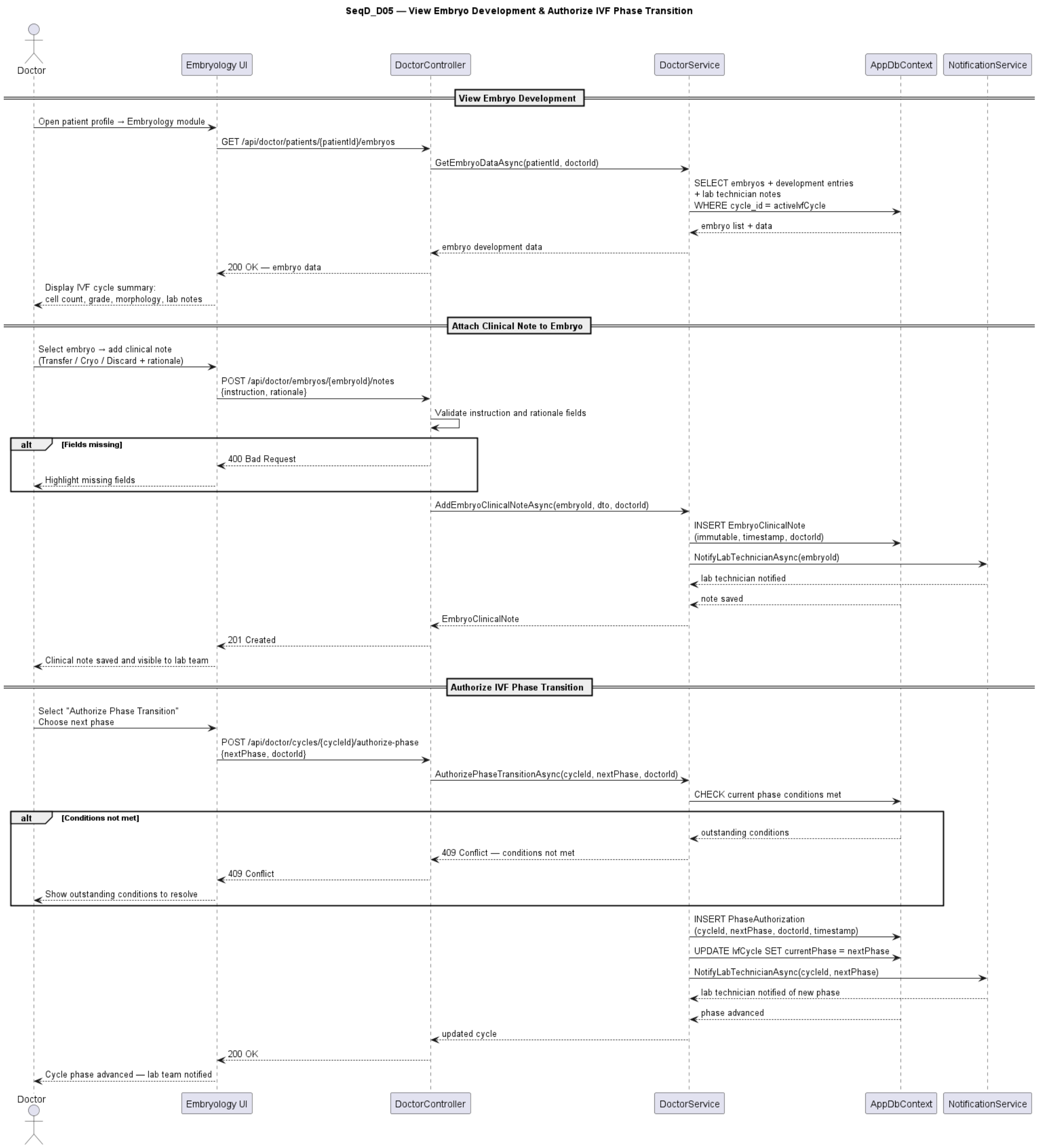
Sequence diagram for UC_D03: Create and Manage Appointments

10.4 UC_D04: Request Lab Test and Review Results



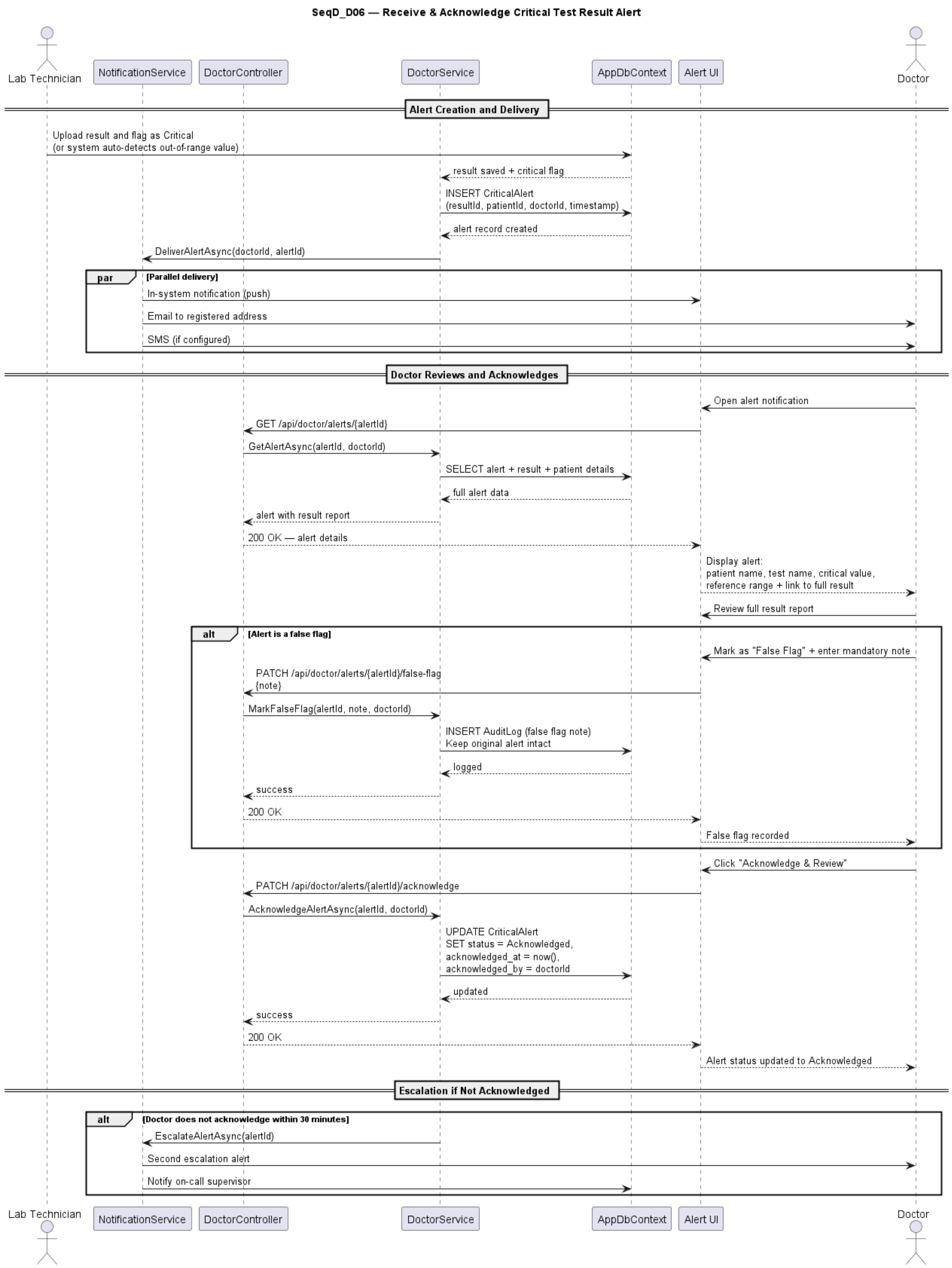
Sequence diagram for UC_D04: Request Lab Test and Review Results

10.5 UC_D05: View Embryo Status and Add Clinical Notes



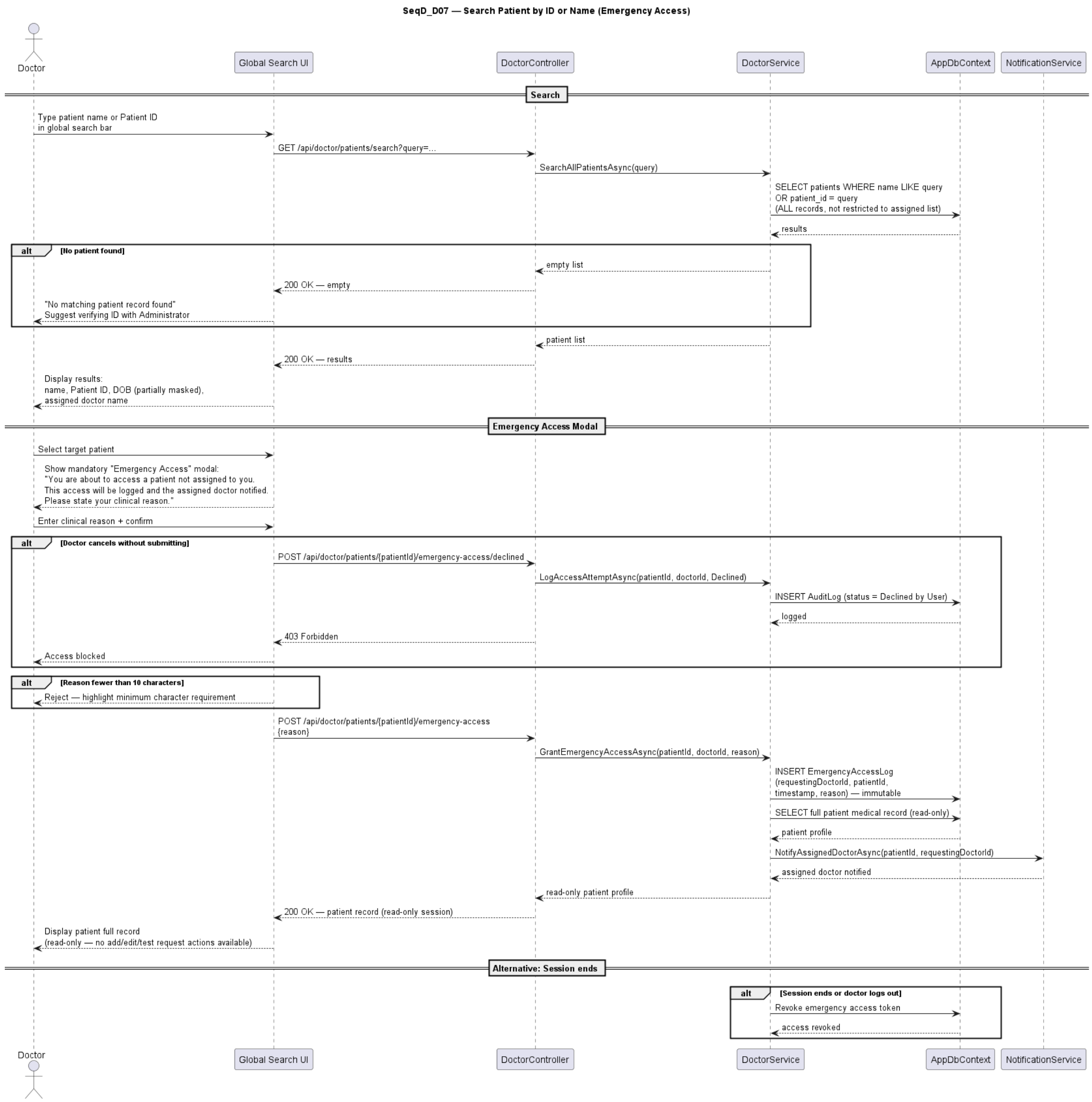
Sequence diagram for UC_D05: View Embryo Status and Add Clinical Notes

10.6 UC_D06: View IVF Cycle History and Authorize Phase Transition



Sequence diagram for UC_D06: View IVF Cycle History and Authorize Phase Transition

10.7 UC_D07: Emergency Patient Search



Sequence diagram for UC_D07: Emergency Patient Search